Anish Goel

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Profile

Mechatronics engineering undergraduate with hands-on experience in designing, building, and testing autonomous robotic systems across underwater, surface, aerial, and medical domains. Strong background in mechanical design, embedded systems, and control, with multiple field-tested prototypes and patented innovations. Seeking roles in robotics R&D and autonomous systems engineering.

Education

Thapar Institute of Engineering and Technology **Aug 2023 – May 2027**
Bachelor of Engineering in Mechatronics(CGPA: 9.52) *Patiala, Punjab*

Relevant Coursework

- Data Structures
- Robotics Engineering
- Automatic Control Systems
- Embedded Systems
- Machine Learning
- CAD/CAM
- Signal Conditioning
- Industrial Automation
- Sensors and Actuators

Projects

Autonomous Underwater Vehicle (AUV) | *Patent No. 202611022081*
- Co-designed a 1.5 m, 25 kg underactuated torpedo-shaped AUV with single-thruster propulsion and compact belt-driven fin actuation for low-drag maneuverability.
- Developed complete 3D structural models and modular chassis enabling rapid assembly and 5 kg payload adaptability.
- Led embedded systems integration and fabrication, packaging onboard computing, sensors and power systems.

Autonomous Surface Vehicle (ASV) | *In Development*
- Designed and developed an autonomous surface vehicle for real-time water quality monitoring and river mapping.
- Designed full mechanical system and 3D models, ensuring stability, waterproofing, and modular sensor integration.
- Implemented full ROS-based autonomy stack, integrating multi-sensor data, navigation, and mission control for autonomous waypoint tracking.

AI-Integrated Tonometry System | *Patent No. 202511115771*
- Co-development of a low-cost intraocular pressure monitoring device aimed at improving early glaucoma detection.
- Designed full mechanical structure and ergonomic housing, optimizing alignment, portability, and clinical usability.
- Integrated sensing and embedded electronics, reducing system cost by $> 80\%$ compared to conventional devices.

Autonomous Aerial Robotics Platform | *Open-Source Prototype*
- Designed and modeled quadcopter chassis, performing thrust-to-weight and structural stability calculations.
- Optimized frame layout for payload integration and autonomous flight testing.

Self-Balancing Robot
- Designed and fabricated a self-balancing robot, including mechanical structure and custom control circuitry.
- Implemented closed-loop stabilization using IMU-based feedback and PID control for rapid balance recovery.

Liquid Metal 3D Printing System | *Patent No. 202511034015, 202511099165*
- Fully designed an industrial liquid metal 3D printer, engineering the furnace–nozzle assembly and CNC-based multi-axis motion system for direct-write metal deposition.
- Engineered controlled molten-metal mixing enabling fabrication of multi-material and composition-graded structures.

Extracurriculars

Society of Intelligent Systems **Feb 2025 – Present**
Project Lead *Patiala, Punjab*
- Led teams of 25+ students in robotics and automation projects
- Managed 4 major interdisciplinary projects, delivering milestones ahead of schedule

Technical Skills

Software: SolidWorks, AutoCAD, MATLAB/Simulink, ROS, Python, C/C++, Creo, Maxsurf, Eagle, Linux
Hardware: Embedded Systems (Arduino, Raspberry Pi), Sensor Integration (IMU, GPS, Sonar, IR), Actuators (Servo, BLDC Motors), 3D Printing, PCB Design

Awards / Certifications

- Gold Medalist 2025 (Rank #1, Highest CGPA (2nd Year), Mechatronics)
- ROBO-AI Industrial Training (2024, 80 hrs): Robotics, Automation, AI (NVIDIA/MSFT/AWS)